

Lecture 2

Review

$\mathbb{R}$ = real numbers, constructed as completion of rationals, i.e. limits exist

Let $\{x_n\}_{n=1}^{\infty} \subset \mathbb{R}$ be a sequence, we say that

$$x_n \rightarrow L \text{ as } n \rightarrow \infty \quad \text{or} \quad \lim_{n \rightarrow \infty} x_n = L$$

if for every $\varepsilon > 0$, $\exists N > 0$ s.t. $\forall n > N$,
 $|x_n - L| < \varepsilon$.

Example. $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$. Let $\varepsilon > 0$. Then for all

$$n > \frac{1}{\varepsilon}, \quad \left| \frac{1}{n} \right| < \varepsilon \quad \square$$

An infinite series takes the form

$$\sum_{k=0}^{\infty} a_k.$$

-The series converges if the sequence

$$S_n := \sum_{k=0}^n a_k$$

converges.

-The series converges absolutely if the

series $\sum_{k=0}^{\infty} |a_k|$ converges.

• absolute convergence $\Rightarrow$ convergence.

Example 1: geometric progression, $a \in (0, 1)$,
partial sums of $\sum_{n=0}^{\infty} a^n \rightarrow \frac{1}{1-a}$

Example 2: $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$

converges absolutely $\forall x \in \mathbb{R}$

Let's check this: Fix $x > 0$. Choose $N > 2x$.

It suffices to show that

$$\sum_{k=N}^{\infty} \frac{x^k}{k!}$$

converges. Observe that for $k \geq N$,

$$\frac{x^k}{k!} = \frac{x^N}{N!} \cdot \frac{x^{k-N} N!}{k!}$$

$$\leq \frac{x^N}{N!} \frac{x^{k-N}}{N^{k-N}}$$

$$\leq \frac{x^N}{N!} \frac{1}{2^{k-N}}$$

$$\leadsto \sum_{k=N}^{\infty} \frac{x^k}{k!} \leq \frac{x^N}{N!} \sum_{k=N}^{\infty} \frac{1}{2^{k-N}} < \infty$$

$\mathbb{R}^n$ and its subsets

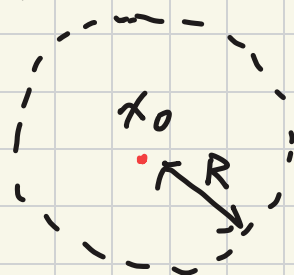
$$\mathbb{R}^n = \left\{ x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} : x_j \in \mathbb{R}, 1 \leq j \leq n \right\}$$

- Dot product: $x \cdot y = \sum_{j=1}^n x_j y_j$
- Euclidean norm: $|x| = \sqrt{x \cdot x} = \sqrt{\sum_{j=1}^n x_j^2}$
- Euclidean distance $|x - y|$
- limit $\lim_{k \rightarrow \infty} x_k = w$ if $\lim_{k \rightarrow \infty} |x_k - w| = 0$

Topology of $\mathbb{R}^n$:

Def. an open ball of radius R centered at $x_0 \in \mathbb{R}^n$ is given by

$$B(x_0; R) := \{x \in \mathbb{R}^n : |x - x_0| < R\}$$



- a neighborhood of x_0 is a small open ball centered at x_0 .
- Let $A \subset \mathbb{R}^n$. If $x \in A$ has a neighborhood contained in A , then x is in the interior of A . If every $x \in A$ is an interior point, then A is called open.

Q1. Finite intersections/unions still open?

Q2. What about infinite?

- $F \subset \mathbb{R}^n$ is called closed if the complement F^c is open

Fact: $\bar{F}$ closed $\Leftrightarrow$ every limit point is contained in F

$\Rightarrow$ intersection of closed sets is closed

- The closure of A is the smallest closed set that contains A i.e.

$$\bar{A} := \bigcap_{\substack{F \text{ closed} \\ A \subset F}} F = \{ \text{limit points of } A \}$$

- The boundary of A is given by

$$\partial A = \{ x \in \mathbb{R}^n : B(x, R) \cap A \neq \emptyset \text{ and } B(x, R) \cap A^c \neq \emptyset \quad \forall R > 0 \}$$

If A is open, then $\partial A = \bar{A} \setminus A$

- set A is bounded if there exists $R > 0$ s.t.

$$A \subset B(x, R)$$

- set $K \subset \mathbb{R}^n$ is compact if it is both closed and bounded

Rmk: from now on, $\Omega \subset \mathbb{R}^n$ will always be an open domain

Functions on $\mathbb{R}^d$

Let $\Omega \subset \mathbb{R}^d$, $f: \Omega \rightarrow \mathbb{R}$

Def. f is continuous at $x_0 \in \Omega$ if for every $\varepsilon > 0$, $\exists \delta$ s.t. if $|x - x_0| < \delta$, then

$$|f(x) - f(x_0)| < \varepsilon$$

Equivalently: $\forall \{x_n\}_{n=1}^{\infty} \subseteq \Omega$ s.t. $x_n \rightarrow x_0$,
 $f(x_n) \rightarrow f(x_0)$

If f is continuous at all $x \in \Omega$, then we say that f is continuous. The set of all continuous functions on Ω is denoted

$$C^0(\Omega) = \{\text{continuous functions on } \Omega\}$$

Let $f_n \in C^0(\Omega)$ be a sequence of fns,

$f_n \rightarrow f$ pointwise if $f_n(x) \rightarrow f(x) \forall x \in \Omega$

$f_n \rightarrow f$ uniformly if $\sup_{x \in \Omega} |f_n(x) - f(x)| \rightarrow 0$

Note uniform convergence $\Rightarrow$ pointwise converg.

Prop. if $f_n \in C^0(\Omega)$ and $f_n \rightarrow f$ uniformly, then $f \in C^0(\Omega)$

Cor. $C^0(\Omega)$ with norm $\|f\|_{C^0} = \sup_{x \in \Omega} |f(x)|$ is complete